

Association of Patient Treatment Preference With Dropout and Clinical Outcomes in Adult Psychosocial Mental Health Interventions

A Systematic Review and Meta-analysis

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 Supplemental content

IMPORTANCE Receiving a preferred treatment has previously been associated with lower dropout rates and better clinical outcomes, but this scenario has not been investigated specifically for psychosocial interventions for patients with a mental health diagnosis.

OBJECTIVE To assess the association of patient treatment preference with dropout and clinical outcomes in adult psychosocial mental health interventions via a systematic review and meta-analysis.

DATA SOURCES The Cochrane Library, Embase, PubMed, PsychINFO, Scopus, Web of Science, Nice HDAS (Healthcare Databases Advanced Search), Google Scholar, BASE (Bielefeld Academic Search Engine), Semantic Scholar, and OpenGrey were searched from inception to July 20, 2018, and updated on June 10, 2019.

STUDY SELECTION Studies were eligible if they (1) were a randomized clinical trial; (2) involved participants older than 18 years; (3) involved participants with mental health diagnoses; (4) reported data from a group of participants who received their preferred treatment and a group who received their nonpreferred treatment or who were not given a choice; and (5) offered at least 1 psychosocial intervention.

DATA EXTRACTION AND SYNTHESIS Two researchers extracted study data for attendance, dropout, and clinical outcomes independently. Both assessed the risk of bias according to the Cochrane tool. Data were pooled using random-effects meta-analyses.

MAIN OUTCOMES AND MEASURES The following 7 outcomes were examined: attendance, dropout, therapeutic alliance, depression and anxiety outcomes, global outcomes, treatment satisfaction, and remission.

RESULTS A total of 7341 articles were identified, with 34 eligible for inclusion. Twenty-nine articles were included in the meta-analyses comprising 5294 participants. Receiving a preferred psychosocial mental health treatment had a medium positive association with dropout rates (relative risk, 0.62; 0.48-0.80; $P < .001$; $I^2 = 44.6\%$) and therapeutic alliance (Cohen $d = 0.48$; 0.15-0.82; $P = .01$; $I^2 = 20.4\%$). There was no evidence of a significant association with other outcomes.

CONCLUSIONS AND RELEVANCE This is the first review, to our knowledge, examining the association of receiving a preferred psychosocial mental health treatment with both engagement and outcomes for patients with a mental health diagnosis. Patients with mental health diagnoses who received their preferred treatment demonstrated a lower dropout rate from treatment and higher therapeutic alliance scores. These findings underline the need to accommodate patient preference in mental health services to maximize treatment uptake and reduce financial costs of premature dropout and disengagement.

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JAMA Psychiatry. 2020;77(3):294-302. doi:10.1001/jamapsychiatry.2019.3750
Published online December 4, 2019.

Patient preference can be defined as the conditions and activities that patients desire in their treatment.¹ International guidelines recommend that clinicians and patients work together to make decisions about treatment,²⁻⁴ fostering a sense of autonomy and empowerment for those with mental health conditions.^{5,6} Involvement in treatment decisions has been associated with higher patient satisfaction,⁷ and accommodation of preferences plays an important role in therapy dropout and clinical outcomes.⁸ Despite these recommendations, patients receiving mental health services continue to report that they want more involvement in treatment decisions.^{9,10} To our knowledge, meta-analyses examining patient preferences to date have included participants seeking treatment for a broad range of physical and mental health concerns but have not yet examined outcomes for patients with a mental health diagnosis.^{8,11,12}

Psychosocial interventions are defined as any psychological or social treatments,¹³ and they form a major component in mental health care. They rely on social interactions for change to occur¹⁴; therefore, regular attendance at sessions and the establishment of a good therapeutic alliance is essential before any clinical benefit may be gained.^{15,16} This scenario is in contrast to the treatment of physical conditions, in which psychosocial interventions are often secondary to treatment of the physical condition itself.

For patients with a mental health diagnosis, engagement with treatment and noncompliance are problematic across both pharmacologic and psychosocial treatments.¹⁷⁻²⁰ It has been estimated that, within psychotherapy, 19% of patients do not complete their treatment,²¹ resulting in poor use of resources and slower rates of recovery.²² Challenges exist in committing to attendance for an agreed time frame, particularly when these commitments are long term.²³ Patients report that opportunities for autonomy, as well as the expected benefits of treatment, facilitate attendance at community therapy groups, whereas lack of information about the treatment on offer is a barrier to attendance.²⁴ The opportunity to choose a preferred treatment may support these processes, thus leading to increased uptake of treatment.

As mental health services face increasing pressure with regard to resources, a better understanding of the factors associated with dropout and clinical outcomes for patients could enable higher efficiency, increased use of services, and financial savings. Previous reviews suggest that accommodation of patient preferences is associated with better clinical outcomes and lower dropout rates.^{8,11,12} However, none of these reviews examined the effect of treatment preference solely for patients with mental health diagnoses.

This review examines a focused sample: participants with diagnosed mental health conditions. It also includes both individual and group treatment, which reflects a mental health services provision. We excluded studies that examined preferences for therapist demographics or preferences about the roles played by the client and therapist²⁵ and focused only on participants who expressed a preference for one treatment over another. Because previous meta-analyses about preference could not be directly associ-

Key Points

Question Is the receipt of a preferred treatment associated with dropout and clinical outcomes in adult psychosocial mental health interventions?

Findings This systematic review and meta-analysis of 29 randomized clinical trials involving 5294 participants with a mental health diagnosis found that receiving preferred psychosocial mental health treatment was associated with lower dropout rates and had a medium positive association with therapeutic alliance. There was no evidence of a significant association with clinical outcomes.

Meaning Offering patients with a mental health diagnosis their preferred treatment is associated with important aspects of engagement in psychosocial interventions; these findings strengthen existing policy and guidance on ensuring informed treatment choice in mental health care.

ated with mental health services, it was deemed important to explore preference accommodation for this client group.

Specific review questions were as follows: (1) What is the association of patient treatment preference with dropout rates and attendance for psychosocial interventions among patients with mental health conditions? (2) What is the association between patient treatment preference and clinical outcomes of psychosocial interventions for patients with mental health conditions?

Methods

This systematic review and meta-analysis was conducted according to a protocol registered on PROSPERO (CRD42018104333) and was reported according to PRISMA guidelines.²⁶ The searches, screening, and extraction were completed by 2 of us (E.W. and H.T.). Agreement between authors was high (87%), and any disputes were resolved through discussion and involving a third author (C.C.) when necessary.

Search Strategy

The search strategy was developed with an information scientist. Between July 16 and 20, 2018, the Cochrane Library, Embase, PubMed, PsychINFO, Scopus, Web of Science, and Nice HDAS (Healthcare Databases Advanced Search) were searched using terms relating to mental health, randomized clinical trials (RCTs), patient preference, and psychosocial interventions (see eAppendix 1 in the [Supplement](#) for full search terms). All articles from the inception of the databases were included. An updated search performed on June 10, 2019, found no further articles suitable for inclusion. Unpublished material was searched on Google Scholar, BASE (Bielefeld Academic Search Engine), Semantic Scholar, and OpenGrey. Translation of articles not in English was planned but not required. The researchers conducted hand searches using reference lists of included articles. Study-level data were sought, and when these data were not reported sufficiently, authors were contacted for clarification.

Study Selection

Studies were eligible if they (1) were an RCT; (2) involved participants older than 18 years; (3) involved participants with mental health diagnoses; (4) reported data comparing a group of participants who received their preferred treatment with a group who received their nonpreferred treatment or who were not given a choice; and (5) offered at least 1 psychosocial intervention.

Data Extraction and Quality Assessment

An extraction framework was piloted by 2 of us (E.W. and H.T.), and the agreed criteria were used to code the included articles in EPPI Reviewer, version 4 (EPPI-Centre, Social Science Research Unit, UCL Institute of Education, University of London).²⁷ The information extracted included country of study, number of participants, mean age of participants, percentage of male participants, percentage of white participants, diagnosis of participants, treatments offered, length of treatment, method of randomization, method of presentation of options, assessment of preferences, follow-up, and outcome measures. The outcomes were extracted for each preference group, rather than for treatment groups. Quality was assessed using the Cochrane Risk of Bias Tool, version 2 (Cochrane).²⁸ The characteristics of studies with insufficient data to be included in the meta-analysis were incorporated in the summaries.

Statistical Analysis

Owing to the high level of clinical and methodological heterogeneity, a random-effects model was used throughout. Continuous data were extracted into Comprehensive Meta-Analysis, version 3 (Biostat Inc).²⁹ All outcome measures were initially considered, and those that were reported by 3 or more studies were included in the final analysis. Given the variance in follow-up time points and in length of treatment, we decided to include pretest and posttest means and SDs to compare those who received their preferred treatment with those who did not or with those who were not given a choice of treatment. If these data were unavailable, mean change scores and SDs, *P* values, or *t* statistics were used. In these cases, the outcome correlation was estimated to be 0.5 after a sensitivity analysis revealed minimal differences when using a correlation coefficient of 0.4 or 0.6.³⁰ A standardized mean difference (Cohen *d*) and SE were calculated for each study within the Comprehensive Meta-Analysis software. When articles reported more than 1 outcome measure for the same symptom, the standardized mean difference and SE were aggregated according to the guidelines developed by Borenstein and colleagues.²⁹ Outcomes were pooled depending on the type of outcome measure, and a random-effects meta-analysis was conducted in Stata, version 15 (StataCorp LLC)³¹ using the standardized mean difference and SE. Pooled effect size for continuous data was interpreted using Cohen *d*, where *d* > 0.2 is small, *d* > 0.4 is medium, and *d* > 0.6 is high.³² For binary dropout and remission rates, raw numbers of events vs no events were recorded and combined to identify a pooled risk ratio using a random-effects meta-analysis in Stata, version 15.³¹

Exploration of Heterogeneity

Heterogeneity was calculated using the *I*² statistic and interpreted per the Cochrane handbook, in which 0% to 40% might not be important, 30% to 40% may represent moderate heterogeneity, 50% to 90% may represent substantial heterogeneity, and 75% to 100% is considerable heterogeneity.³³ Publication bias was assessed by inspecting funnel plots for asymmetry and applying the Egger test for small study effects. When more than 10 studies were combined, heterogeneity was explored further using subgroup analysis of (1) randomization methods, (2) type of treatment (medication as treatment option vs psychosocial treatments only), (3) definition of dropout, and (4) whether the treatment was by group or individual. Meta-regression was conducted to examine the association with percentage of dropouts in the study as well as demographic characteristics of mean age, percentage of participants who were male, and percentage of participants who were white.

Sensitivity Analysis

In several studies, participants were given their choice of treatments if they refused to be randomized. This approach could be seen to introduce significant randomization bias,³⁴ so a sensitivity analysis was conducted excluding these trials.^{35–40}

Results

Selection, Inclusion, and Characteristics of Studies

Overall, 7341 articles were screened, leaving 34 eligible articles to be included in the review (Figure 1). Five studies were not eligible to be included in the meta-analysis but were included in the summary of study characteristics. These studies did not report sufficient outcome data associated with the participants' preference match, and it was not possible to contact the authors.^{41–45}

Most studies were based in the United States.^{37,42,44,46–61} Others were conducted in the United Kingdom,^{40,41,45,62,63} the Netherlands,^{35,38,39} Germany,^{64,65} New Zealand,⁶⁶ Canada,³⁶ Australia,⁶⁷ Italy,⁶⁸ and Iran.⁴³ Most studies focused on participants who received a diagnosis of depression or anxiety (*n* = 24). Others involved participants with alcohol or substance use disorders (*n* = 6), 1 study examined comorbid depression and substance use (*n* = 1), and the remainder (*n* = 3) explored other psychiatric conditions (schizophrenia, personality disorder, or mixed diagnoses). The most prevalent psychosocial treatments were cognitive and/or behavioral therapy (*n* = 18), counseling (*n* = 4), and interpersonal therapy (*n* = 4). Twenty-one studies offered medication as one of the treatment options. The number of treatment sessions offered varied from 9 to 52. A full list of treatments are presented in eAppendix 2 in the Supplement.

Studies used the following 3 main methodological approaches: (1) asking participants their preferred treatment and then fully randomizing participants to treatment groups (*n* = 20); (2) randomizing participants to either a choice arm, in which they received their preferred treatment, or a no-choice arm, in which they were then rerandomized into a treat-

ment group ($n = 7$); and (3) allowing participants who refused to be randomized or who were disengaged to receive their preference, with all other participants being randomized ($n = 6$). The outcomes of those who received a preferred treatment or who were given a choice of treatment were grouped together in the analysis and compared with those who received a non-preferred treatment or were not given a choice of treatment.

Most studies did not report how they informed participants of the treatment options ($n = 24$). Others used a written description ($n = 6$) or verbal script or discussion ($n = 3$), and 1 study showed participants a video. Fifteen studies did not report their method for assessing participants' treatment preference, 14 used a rating scale or questionnaire (7 of which were validated and the remainder researcher designed), and 5 ascertained preferences through discussion with participants.

Results of a risk of bias assessment (eAppendix 3 in the Supplement) implied that blinding of participants and clinicians was not usually possible; therefore, 29 studies (85%) were deemed to be high risk, with the remaining 5 studies (15%) rated as unclear. A total of 19 studies (56%) reported the blinding of assessors, and 28 studies (82%) reported adequate random-sequence generation for participants who were randomized. A total of 8 studies (24%) reported only a subselection of predefined outcome measures associated with preference and were deemed to demonstrate reporting bias, and 7 studies (21%) reported high dropout rates, which would lead to incomplete outcome data.

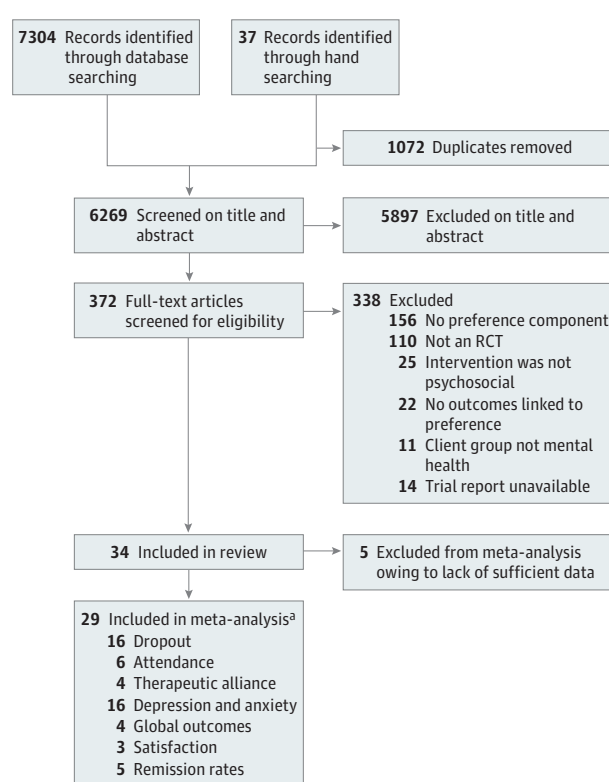
The 29 studies eligible to be included in the meta-analysis were grouped according to the outcome measures used. Depression and anxiety outcomes were combined owing to their clinical similarity as common mental health problems.⁶⁹ Data on addiction severity, substance misuse, schizophrenia symptoms, and treatment response rates were excluded because fewer than 3 studies reported these outcomes. A summary of meta-analysis results for each separate outcome can be found in the Table.

Results of the Meta-analyses

Full results are summarized in the Table. Mean attendance (eAppendix 4 in the Supplement) was reported by 6 studies and was not associated with patient preference ($d = 0.37$; 95% CI, -0.13 to 0.39 ; $P = .33$; $I^2 = 56.8\%$). Dropout rates were reported by 16 studies involving 1857 participants (eAppendix 5 in the Supplement). A significant pooled risk ratio of 0.62 (95% CI, 0.48 – 0.80 ; $P < .001$; $I^2 = 44.6\%$) was obtained, implying a positive association with preference accommodation (Figure 2). Of 846 participants who received their preferred treatment, 159 (18.8%) dropped out; of 1011 participants who did not receive their treatment preference or were not offered a choice, 339 (33.5%) dropped out.

Therapeutic alliance (eAppendix 6 in the Supplement), reported by 4 studies with a total of 199 participants, was found to be significantly stronger for participants who received their preferred treatment ($d = 0.48$; 95% CI, 0.15 – 0.82 ; $P = .01$; $I^2 = 20.4\%$) (Figure 3). Depression and anxiety symptoms were reported by a total of 16 studies, with 3972 participants (eAppendix 7 in the Supplement). There was no evidence that being matched with a preferred treatment was associated with im-

Figure 1. PRISMA Flow Diagram



RCT indicates randomized clinical trial.

^a Some studies reported data that were included in more than 1 of the meta-analyses.

proved depression and anxiety symptoms ($d = 0.01$; 95% CI, -0.18 to 0.20 ; $P = .91$; $I^2 = 83.9\%$). Global outcomes, including the Global Assessment Scale and Clinical Global Impressions Scale, were reported in 4 studies with a total of 686 participants (eAppendix 8 in the Supplement), and they did not show evidence of being associated with preference accommodation ($d = 0.15$; 95% CI, -0.04 to 0.33 ; $P = .12$; $I^2 = 12.6\%$). Satisfaction with treatment was reported by 3 studies, with a total of 1983 participants (eAppendix 9 in the Supplement). No significant association between treatment preference and satisfaction with treatment was found ($d = -0.03$; 95% CI, -0.12 to 0.06 ; $P = .55$; $I^2 = 0\%$). Five studies were included in the assessment of clinical remission rates, with 813 participants (eAppendix 10 in the Supplement); no association between treatment preference accommodation and clinical remission rates was found (risk ratio, 1.03; 95% CI, 0.69 – 1.55 ; $P = .88$; $I^2 = 80.4\%$).

Heterogeneity

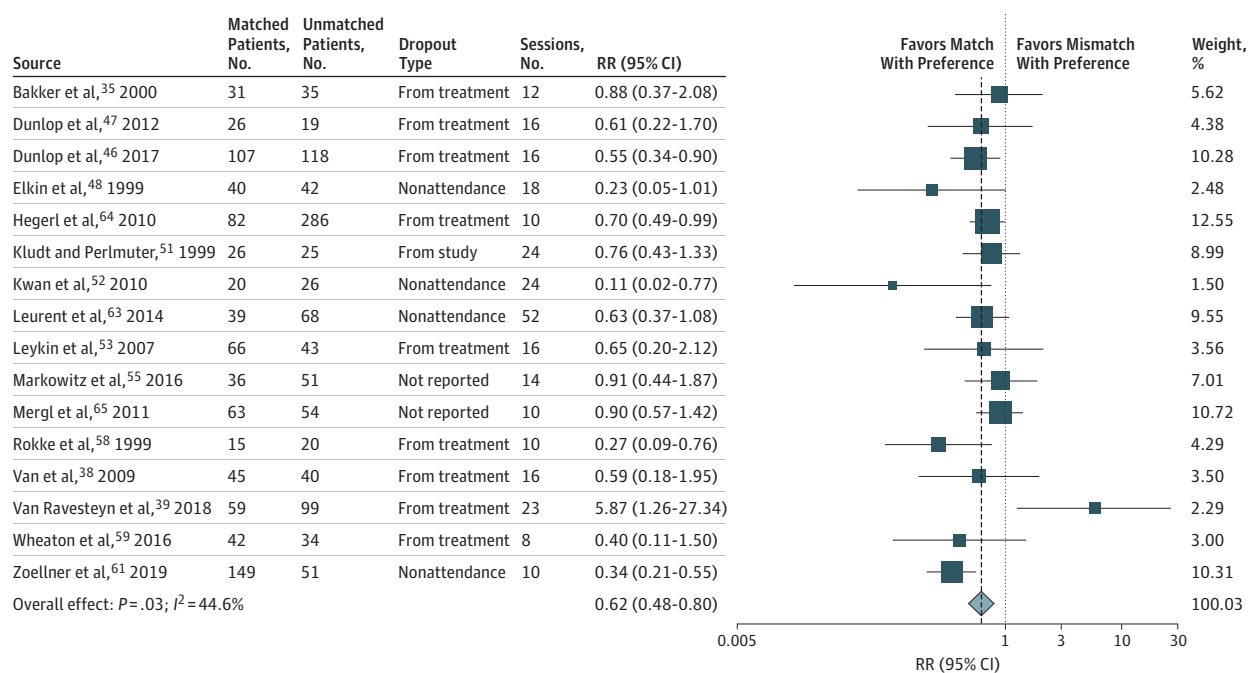
In subgroup analysis of dropout rates (eAppendix 11 in the Supplement), studies that offered medication as an option were not significantly heterogeneous, nor were studies that randomized participants after their preferences were elicited. When examining definitions of dropout, heterogeneity was no longer significant within groups. For depression and anxiety

Table. Summary of Results of Meta-analyses

Outcome	No. of Studies	Participants in Each Group, No.		Effect Size (95% CI)	P Value	P Value		
		Matched to Treatment Preference	Unmatched to Treatment Preference			I^2 , %	I^2	Egger Test
Attendance at sessions	6	287	292	Cohen d , 0.37 (−0.13 to 0.39)	.33	56.8	.04	.51
Dropout	16	846	1011	RR, 0.62 (0.48 to 0.80)	<.001	44.6	.03	.38
Therapeutic alliance	4	95	104	Cohen d , 0.48 (0.15 to 0.82)	.01	20.4	.29	.39
Depression and anxiety	16	2100	1872	Cohen d , 0.01 (−0.18 to 0.20)	.91	83.9	<.001	.94
Global outcomes	4	289	397	Cohen d , 0.15 (−0.04 to 0.33)	.12	12.6	.33	.81
Satisfaction with treatment	3	1142	841	Cohen d , −0.03 (−0.12 to 0.06)	.55	0	.63	.97
Remission rates	5	526	287	RR, 1.03 (0.69 to 1.55)	.88	80.4	<.001	.51

Abbreviation: RR, risk ratio.

Figure 2. Forest Plot of Dropout Rates



Weights are from random-effects analysis. The dashed line indicates the overall risk ratio (RR). The size of the squares is proportional to the study's weight in the meta-analysis. The diamond represents the overall RR, with the edges of the diamond being the 95% CI.

measures, considerable heterogeneity between studies was observed, which was not accounted for through subgroup analysis or metaregression for dropout rates (eAppendix 11 in the [Supplement](#)). Metaregression of demographic characteristics found no significant interaction of mean age, percentage of participants who were male, or percentage of participants who were white with either anxiety and depression outcomes or dropout rates (eAppendix 11 in the [Supplement](#)).

Sensitivity Analysis

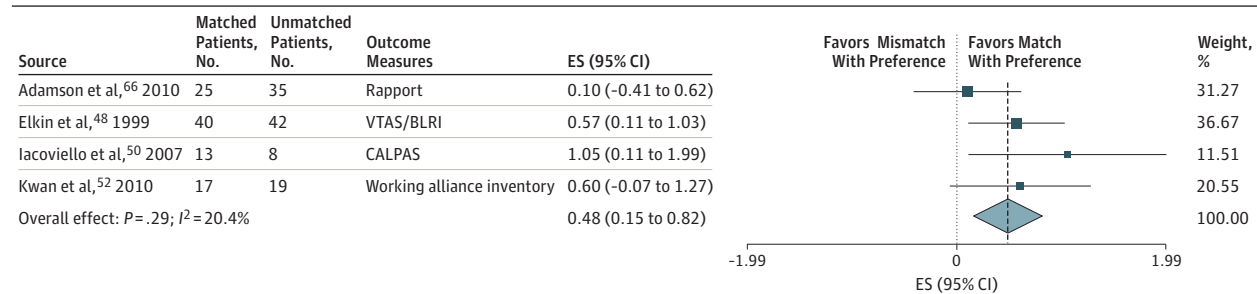
Funnel plots did not show evidence of publication bias, and the Egger test confirmed this outcome. Studies using the

method in which those who refused to be randomized were given their treatment preference were removed in a sensitivity analysis, which did not change the significance of any meta-analysis result.

Discussion

To our knowledge, this is the first meta-analysis to examine associations of patient preference accommodation with dropout and clinical outcomes of patients with a mental health diagnosis. We found that receiving a preferred psychosocial in-

Figure 3. Forest Plot of Therapeutic Alliance



The dashed line indicates the overall effect size (ES). BLRI indicates Barret-Lennard Relationship Inventory; CALPAS, California Psychotherapy Alliance Scale; and VTAS, Vanderbilt Therapeutic Alliance Scale. The size of the squares is proportional to the study's weight in the meta-analysis. The diamond represents the overall ES, with the edges of the diamond being the 95% CI.

intervention was associated with significantly lower dropout rates and a significantly stronger therapeutic alliance. No association was found between receiving a preferred treatment and depression and anxiety outcomes, global outcomes, attendance, or remission rates. To our knowledge, this is a new and relevant finding for the field of mental health research and service provision.

This meta-analysis confirms findings from previous meta-analyses that preference accommodation is associated with lower dropout rates, but it differs in that no significant association was found with clinical outcomes or treatment satisfaction.^{8,12,70,71} Our findings support the suggestion of Lindhiem et al¹² that therapeutic alliance is improved when patients receive a preferred treatment. A strong therapeutic alliance has also previously been associated with reduced dropout rates,⁷² which adds to the understanding of our findings.

It was not possible to generalize findings from previous meta-analyses to inform mental health services or research; therefore, this study focused only on patients with mental health diagnoses. In the present study, we grouped outcomes of a similar nature, whereas previous meta-analyses used outcome as a moderator analysis.⁸ We also used a more cautious method of combining outcome effects from the same study, which could have led to smaller effect sizes and therefore a non-significant association for depression and anxiety outcomes.⁷³ The present study examined a client group with diagnosed mental illness; therefore, it is possible that when a condition is more severe, preference accommodation may have a weaker association with clinical outcomes. In the study by Swift et al,⁸ a stronger association of preference at a midtreatment time point was found. In the present study, narrower inclusion criteria meant that there were limited data to examine midtreatment effects. The wide variance in follow-up time points meant that postintervention data were more reliably available.

The most robust association of preference across client groups is with dropout rates.^{8,12,70} Few studies gave a clear definition of dropout in the present review, with little agreement among those that did. This variation may partly explain the heterogeneity observed in these studies. In addition, dropout from individual therapy has been found to be higher when there are more sessions offered.²³ The range of the number of

sessions offered (9-52) in this review could therefore also account for some of the heterogeneity. When patients drop out of treatment early, it is likely to have a negative association with the efficacy of the treatment.¹⁷ Therefore, accommodation of preferences is an important consideration for all mental health service professionals and researchers in the field of mental health.

It has been debated whether RCTs are a reliable method of assessing the effect of preference.^{34,74,75} Patients who have strong preferences may be unwilling to enter into an RCT in which they could be allocated to their nonpreferred treatment. Some studies in our review attempted to address this issue by giving patients a choice if they refused to be randomized. However, this approach can lead to overinflated effect sizes and is not recommended.⁷⁶ In our analysis, studies that did not fully randomize their participants were removed in a sensitivity analysis, but they were not found to be associated with the results.

It has been suggested that participants may be more willing to be randomized in trials, even when they have a strong preexisting treatment preference, if they are given comprehensive information about each treatment option and there are no substantial differences in the efficacy of treatment options.³⁴ Despite this possibility, few articles reported how they presented the available treatment options to participants. Although it has been previously found that informed vs uninformed choice is not a moderator when exploring preference accommodation,¹² transparency about how information is presented to participants is required for both replicability and a greater understanding of how this transparency may be associated with their choices. It has also been recommended that measurements of preference should be assessed for validity⁷⁷; however, most studies included in this review did not report their methods of measuring preferences.

Limitations

This study must be interpreted with consideration of its limitations. Despite our intentions to combine only similar outcomes, this study still pooled varying measures, which may compromise the accuracy of the findings. Significant heterogeneity was present in many of the analyses, which could influence comparability between studies. However, these dif-

ferences would have been accounted for to some extent through the use of a random-effects meta-analysis.⁷⁸ Studies varied in their design but were clinically comparable owing to the similarities in participants' diagnoses, the treatments that were offered, and the experience of being given a preferred treatment or being given a choice of treatment. The included studies were situated mostly in the United States and mostly among participants with depression. More evidence should be gathered about the associations between preference accommodation and outcomes in different countries, with varied health care systems, and with participants with a wider range of diagnoses before the results can be generalized.

This study examined the differences in outcomes between patients who did and patients who did not express a preference for the given treatment. Within the RCT design, patients were randomized to treatment groups or to choice groups, giving a random distribution of potentially confounding factors across treatment and preference groups. The removal of studies that did not fully randomize in a sensitivity analysis highlighted that these studies showed that the results of the meta-analysis were stable against this criterion. An alternative approach would be to conduct an individual participant data meta-analysis to explore preference associations further, including interactions between preference, patient characteristics, and the treatment received. A propensity score could be calculated to reduce bias between the nonrandomized groups,⁷⁹ but owing to the limited availability of this level of data, this approach was not possible for the present study.

Recommendations

The current findings have implications for the way in which services offer treatment. Even if the range of available psy-

chosocial interventions is limited within services, this study suggests that offering a basic level of choice (eg, medication or psychotherapy) can be beneficial to both patients with mental health diagnoses and mental health services. Although it may not be directly associated with treatment outcomes, receiving a preferred treatment is associated with a better therapeutic alliance and lower dropout rates. Engagement with treatment is crucial for meaningful change to take place, and therapeutic alliance is seen as an essential mechanism for change in psychotherapy,¹⁴ with robust evidence for its positive association with one-to-one treatment outcomes.⁸⁰

Conclusions

Receipt of a preferred treatment is associated with lower dropout rates and a stronger therapeutic alliance for patients with mental health diagnoses. These findings are valuable for clinicians, mental health services, and researchers when offering treatment to patients with mental health diagnoses. Mental health services can increase engagement with psychosocial interventions by giving patients opportunities to make informed choices about their care, which is in line with current guidelines and practice. Given the lack of clarity as to how treatment options were presented or measured, full reporting is recommended for future studies to enable translation into clinical practice. For future investigations of patient preference, it is recommended that individual patient data sets be examined to make preference groups comparable. For this method to be possible, RCT publications must make their full data sets readily available.

ARTICLE INFORMATION

Accepted for Publication: September 24, 2019.

Published Online: December 4, 2019.

doi:10.1001/jamapsychiatry.2019.3750

Author Contributions: Ms Windle had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Windle, Priebe, Carr.
Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: Windle, Carr.
Critical revision of the manuscript for important intellectual content: All authors.

Statistical analysis: Windle, Sabitova.

Obtained funding: Priebe.

Administrative, technical, or material support:

Windle, Tee, Sabitova, Priebe, Carr.

Supervision: Jovanovic, Priebe, Carr.

Conflict of Interest Disclosures: None reported.

Funding/Support: This review was sponsored by the East London National Health Service Foundation as part of a PhD studentship.

Role of the Funder/Sponsor: The funding source had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

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